

## Tastants

Small particle taste  
Salty or sour  
Perceived via G protein-coupled receptors, taste sweet, bitter or umami

Salty: presence of sodium ions, entry of  $Ca^{2+}$  via voltage gated calcium channel

Sour: activate by Taste  
hydrogen ion via hydrogen ion channels

Fats: a receptor responds to fatty acids

Sweet: receptor are member of G-protein coupled receptor

Bitter:

Umami: Glutamate receptor and amino acids

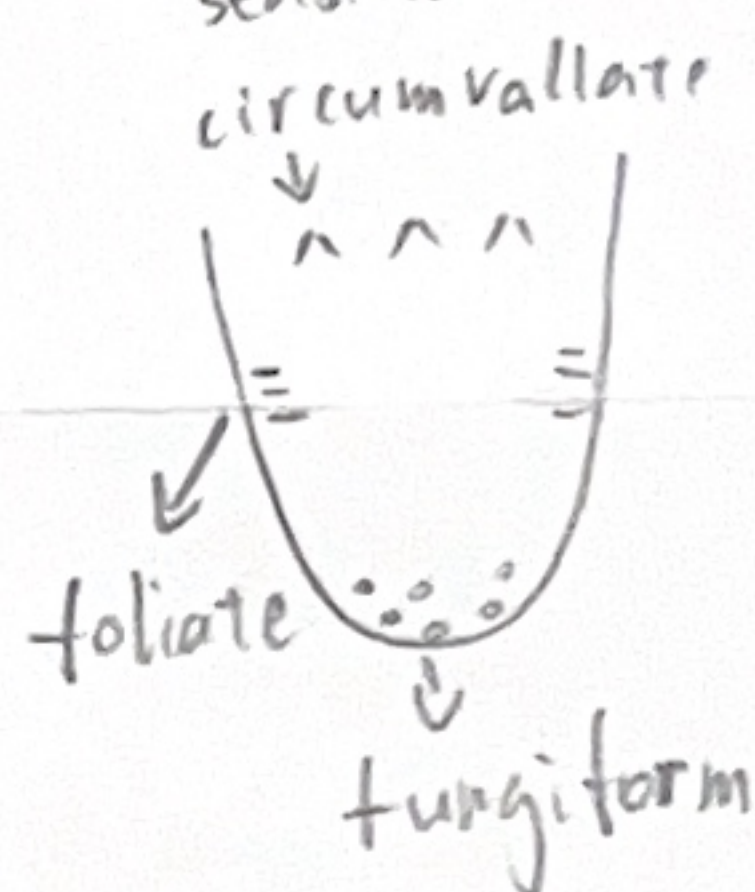
## Papillae

① Filiform PP  
→ small structure on tongue no taste

② Fungiform PP  
→ mushroom shape on edge of tongue tip.  
six taste buds per PP

③ Foliate papillae  
→ on rear of the tongue

④ Circumvallate PP  
→ inverted V-shape on rear of the tongue  
larger than ② have less taste buds, sensitive



## Detect smell

→ Olfactory membranes which molecule must penetrate

→ 200 distinct nasal receptors, the cilia receptors capture scent molecules.

## Test

Preference: two-choice or multiple choice

Descriptive: long question

Discrimination:

Pair compare: see if two product differ

Triangle: two same one different

## Off-flavours

### ① 2-Acetylpyridine

- form during malt kilning
- biscuit or corn chip

### ③ Isoamyl acetate

- form when alcohol react with acetic acid (vinegar)
- banana, pear

### ⑤ Isobutyraldehyde

- from husks of grains when sparging
- brown sugar, cheap ice cream

### ② 4-vinyl guaiacol

- form by enzyme released by yeast
- spicy, clove-like

### ④ Geraniol

- from hop oil
- flowery, rose

### ⑥ Dimethyl Sulphide (DMS)

- form when heating malt
- cooked corn, buttery

\* Ester: fruity

\* Nucleation: formation of bubbles in a carbonated drink

### ⑦ 3-methyl-2-butene-1-thiol

- form by light strike
- skunked

Oxidation: loss of electrons

Antioxidants prevent oxidation

- Hops have an antioxidant call Xanthohumol

Flavour of Oxidation

- Trans-2-nonenal: wet cardboard
- Benzaldehyde
- Diacetyl 1-2-3-pentanedione

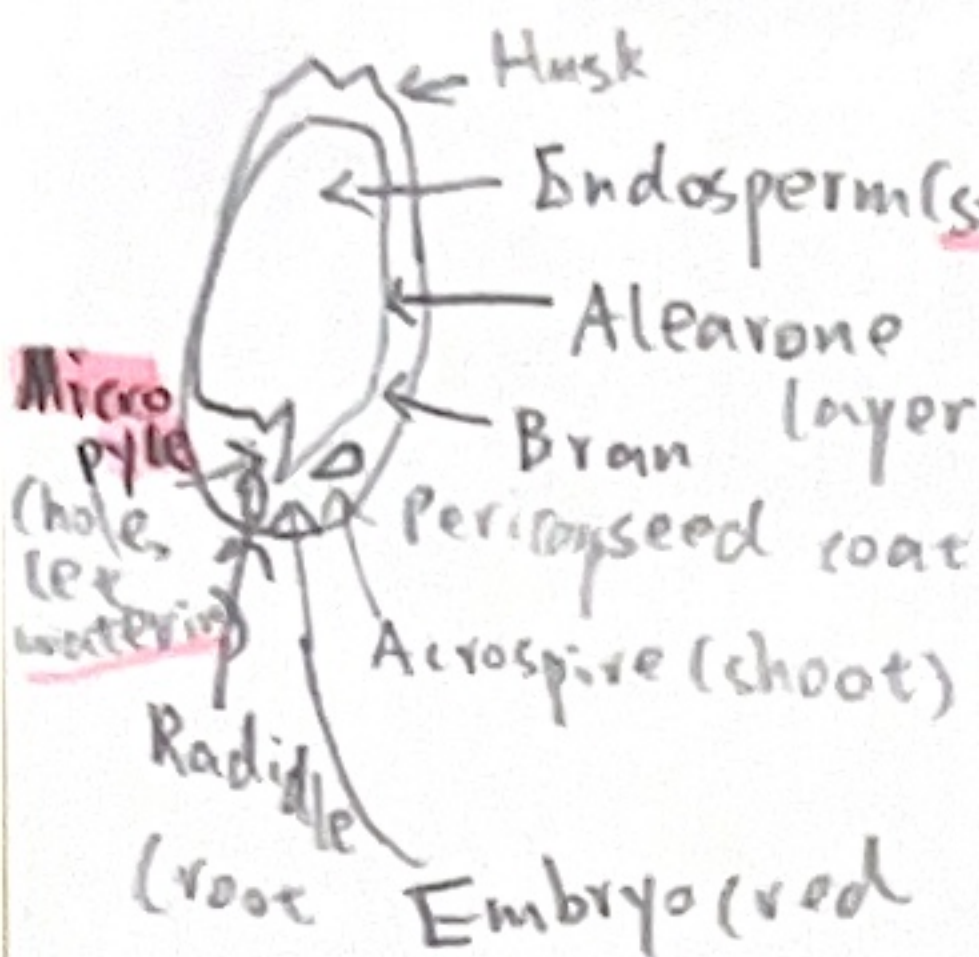
Cause

- Time
- Temperature increase rate of oxidation
- Light → loss of electrons from molecules

## Barley

animal food (75%) malt for (20%) brewing human food (5%)

- 1. Six-row barley
  - higher nitrogen, enzymatic content
  - smaller grain size
  - American
- 2. Two-row
  - British, European, Australian & New Zealand



- Barley is preferred as
- 1) has protective husk
  - 2) 90% are carbohydrate
  - 3) aleurone layer is thick
    - ↓ produces enzymes
  - 4) gelatinisation temp is low (52-59°C) 糊化温度

Plant Breeders Rights: a type of copyright which protects the plant breeder's IP for unique plant varieties

- allow breeder to limit a variety to a grower

End point Royalty (EPR): fee paid on every tonne of grain of a particular variety sold by breeder/grower.

## Hops

- dioecious, (either male or female)
- hop cone covered in ~~stiff~~ bracts and bractioles.
- **lupulin glands** contain hop oils and resins.
- stalk or string is source of polyphenols (tannins)
- female hop cones are used in brewing
- In oast house, dry from 80% to 10% moisture.

### German and Czech hops

- Hallertauer Mittelfruh (Bavarian)
- Saaz

### English hops

- East Kent Golding
- Fuggle

### American hops

- Cascade
- Chinook
- Crystal

### Australian hops

- Pride of Ringwood
- Fuggle x wild Tasmanian
- Ella
- Galaxy

### New Zealand hops

- Nelson Savin

## Water

### Surface water

- weakly acidic water
- picks up organic matter and some minerals.

### Groundwater

- can become highly mineralised and alkaline

### Brewing water

- 1) Precipitation [rainwater]
- 2) Surface water [river]
- 3) Groundwater [well]
- 4) municipal water supply 市政供水

### Temporary hardness

- cause by calcium/magnesium bicarbonate
- boiled water precipitates calcium carbonate

### Permanent hardness

- other calcium & magnesium salts
- not removed by boiling

### Water treatment

- Sedimentation & filtration: remove particles & organic material
- Boiling: remove temporary hardness & Chlorine
- filtration through activated carbon filter: x organic compounds & Chlorine
- Ion exchange resins & reverse osmosis: salt water with applied pressure to pure water

## Bacteria

→ Prokaryotes 原核生物

## Yeast

### ★ Unicellular fungi

★ Saccharomyces cerevisiae

★ Sugar fungus / require nitrogen

★ 通常是无性繁殖 Asexual reproduction

有性繁殖 means it runs out of water/sugar

★ ferment at 18-24°C: 厌氧 Anaerobic condition.

★ fermentation by product: esters, diacetyl

## Malting

the process of soaking grains and allow them to germinate

Purpose: to transform and mobilise the starch in grain to provide a substrate suitable for mashing

Before malting: 弄麦芽前需先达标

1. Germination rate ( $>95\%$ )
  - Protein content (9.5-12.5%)
2. dried (213.5% moisture)

## Malt Quality Parameter

### ① Malt extract

- Malt extract measures the amount of fermentable sugars from a tonne of grain

### ② Diastatic power

- measures the amount of diastatic enzymes.
- low diastatic power associate with low potential for malt extract

### ③ Wort viscosity

- measures gumminess of wort relative to water

## Type of Males

### Pale malts

- 5-10 °EBC

### Dark Malts

- 1000-1200 °EBC

### Amber Malts

- kiln at 150-180°C
- low/no diastatic power
- in English IPA, brown ales

### Caramel/Crystal

- 50-320 °EBC
- improve head stability

## Other malts

### Biscuit malt

- lightly roasted malt

### Acidulated malts

- in sour beers

### Wheat malt

- high diastatic power & protein content
- 80% of German wheat beer

### Smoked malts

- smoked meat flavour

## Malting process

Steeping

12%  $\rightarrow$  45%  $H_2O$

green malts

Germination  $\rightarrow$  Kilning

45%  $H_2O \rightarrow$  4%  $H_2O$

### 1. Steeping

- wet stand (8 hours) dry stand (10-14 hr)
  - ↓ remove  $CO_2$  & control heat
- 2-3 steeping cycles

### 2. Germination

- 10-16°C for 3-5 days
- rehydrated **embryo** releases gibberellic acid (GA3) which ~~se~~ activate ~~er~~ cells of aleurone to release enzyme.
- enzyme function to break down cell wall of **proteinases** and **Beta-glucanases** and release starch.

### 3. Kilning modification (主要)

- dried for 16-60 hours.

### A. Roasting

- Amber (40-80 °EBC)  $\rightarrow$  colour
- Chocolate (1000-1200 °EBC)
- Black malt (1200-1400 °EBC)

### ★ Crystal & Caramel Malts

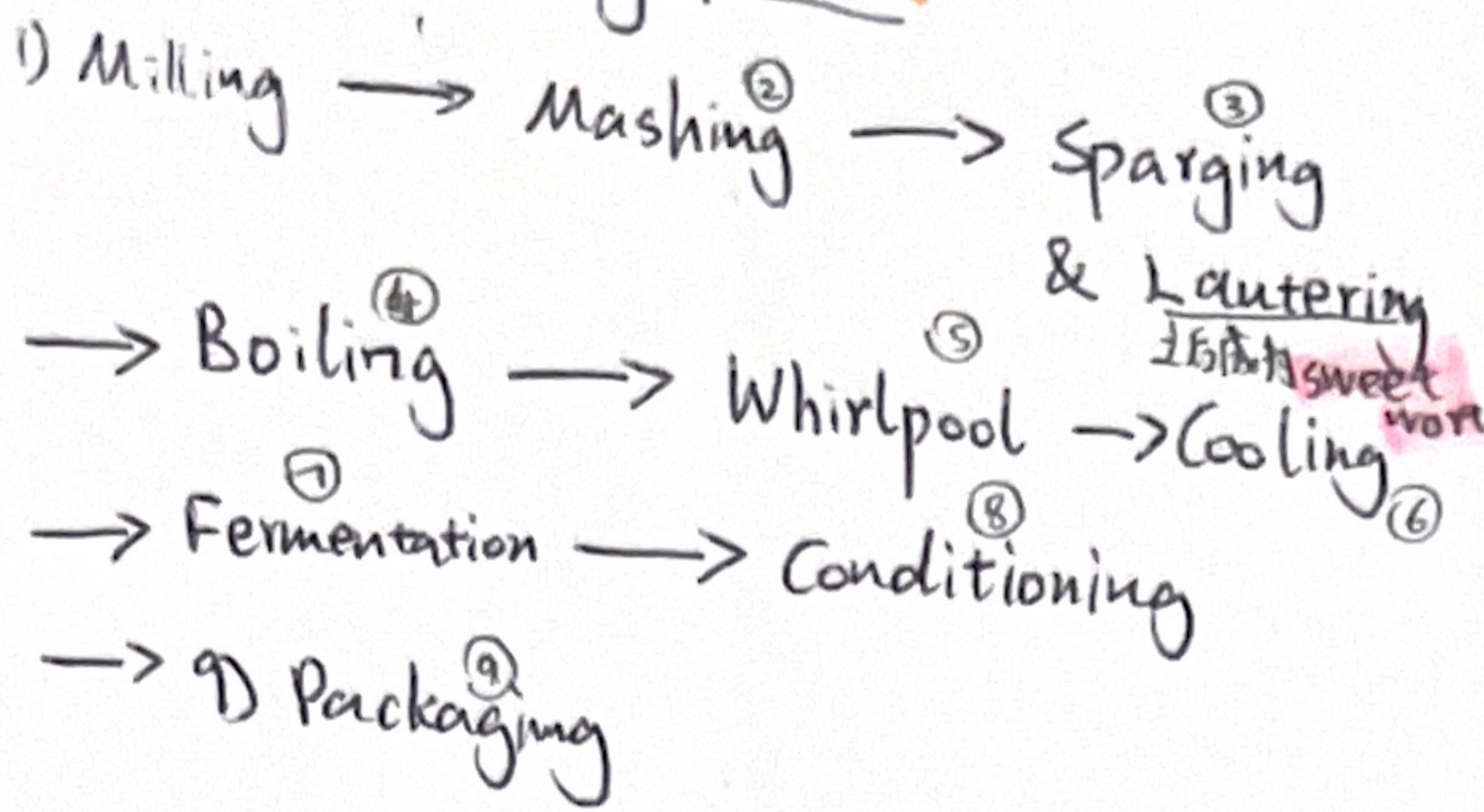
- produced by roasting unkilned green malt (45% moisture)
- **Green malt** is 'stewed' at 55-70°C for 30-40 minute prior to ~~roasting~~ roasting

## Adjuncts

Purpose

1. reduce cost
  - sugar is cheaper than hop
2. increase strength
3. provide quality
  - such as better foaming or head retention

## Brewing process



Aflatoxins 黄曲霉毒素  
• cause hepatic cancer  
• limit of 15 µg/kg for total aflatoxins

Grist = crushed malt  
Husk Grist Floor  
30% 64% 6%

② Mashing  
Saccharification rest  
• 64-67°C  
• pH 5.2-5.7

- Yeast can only use Glucose, maltose, maltotriose, galactose, fructose
- Iodine test  
Black (still have starch in mash X)  
Yellow (conversion done) & sweet wort

### ③ Sparging

- hot water (77°C) sprinkle over grain

### ④ Boiling

- boil for 60-90 minutes

Method: direct heat (hard to clean) (uneven heat) → change colour & aroma & taste  
Steam (uneven heat)  
External calandria ← chimney & spread plate steam heat exchanger Thermosiphon

- purpose of boiling
  - sterilising
  - Inactivate enzymes and coagulate proteins
  - concentrate wort
  - boil off unwanted compounds and dissolve wanted compounds

### Kettle finings

- help precipitate proteins and suspend materials.

### Wort Clarity

- Fining addition
- wort pH
- malt variety & modification
- gravity (糖量)
- mashing schedule
- Boil vigour 活力

### Wort Quality

- Protein → haze & flavour problems
- Tannins → chill haze and astringency
- Lipids → destroy foam
- Spent hops → smother yeast 窒息

### ⑤ Whirlpooling

- hop debris removal
- by spinning of the wort

### ⑥ Cooling

- Purpose: prevent bacteria growth (26°C)
- prevent oxidation
- down to fermentation temperature

Method: coolships  
trickle coolers  
Heat exchanger

occurs below 70°C  
forms 20-30°C

- Cool break: 27°C to 20-30°C  
formed  
improve by kettle finings.

### ⑦ Fermentation needs

- hopped wort
- oxygen
- < 30° to prevent oxidation
- Sugar
- Nitrogen
- Minerals and Vitamins

### Type of Yeast

- Dried Yeast
- Recovered yeast
- Yeast propagation 繁殖

Pitching rate: no. of yeast cells per volume of wort  
Ale 5-15 million cell/ml  
Lager 15-20

### Primary fermentation

- release carbon dioxide
- Ale 5 days 16-20°C
- Lager 21 days 8-14°C
- After done, pH of the green beer should drop to 4.0 and gravity will drop and lower O<sub>2</sub> and higher CO<sub>2</sub>

### Warm maturation

- Purpose: allow yeast convert remain sugar and metabolise by-product
- Ale 1-4 days 12-18°C
- Lager 4 days - 20-26°C few weeks

### ⑧ Cold Conditioning (Lagering)

- Purpose:
  - reduce haze
  - reduce Oxygen
  - Carbonation
  - Stock holding
  - Blending
  - Process aid reaction time
- Ale 1-4 days -1°C
- Lager few weeks -1°C

### ⑨ Packaging & Carbonation

- Carbonation can occur in cask or bottle conditioning.

## Real Ale

- matured by secondary fermentation
- Brewed with **traditional ingredient** grains <sup>↑</sup> yeast <sup>↑</sup> water
- know as "cask-conditioned" beer

## Pale malt

- Before 18th century, kiln over wooden fire
- coal - contain sulphur
- coke - purified coal

## Beer clarity

- ① chill haze (soluble protein)
- ② unseparated yeast
- ③ undegraded starch
- ④ Bacterial growth

## Beer foam

- Improve: protein  
↓ pH & hop compounds  
↑ Nitrogen gas
- Worsen: \* Fats & oils  
\* Detergent

## Brettanomyces spp

"British fungi"

- smell of horse blanket
- volatile phenolic compounds
  - 4-vinyl guaiacol
  - & 4-ethyl phenol
- a positive flavour in some Belgian styles

## Wheat beer

- △ contain high level phenolic compounds
- △ brew with ale yeast
- △ contain high level of esters

## German & Czech Beers

### 1. Lagers:

- Pilsner
  - Dunkel
  - Rauchbier
  - Schwartzbier
  - Bock
  - Helles
  - Vienna / Oktoberfest / Märzen (March beer)
- barley wine, brewed traditionally in October*

### 2. Ales:

- Ales
  - Kölsch
  - Wheat beers
- Brew with Ale yeast at lager yeast temperature

## British Yeast file

- Esters  $\leftarrow$  Isoamyl acetate  
Ethyl acetate
- Phenolics  $\leftarrow$  4-vinyl guaiacol  
4-ethyl phenol
- VDKs - Diacetyl
- Aldehydes - acetaldehyde
- Sulphur  $\leftarrow$  Dimethyl sulphide (DMS)  
CDMS
- Low level in British malts
- \* British beer  
特点: Esters, low Diacetyl

## Terpenoids

### Monoterpenoids (C10)

- menthol
- Eucalyptol
- Citronellol

### Diterpenes (C20)

- Tetrahydrocannabinol
- Steriol
- Retinol

### Tetraterpenes (C40)

- $\alpha$  &  $\beta$
- Lycopene

an aromatic hop oil

## Types of Beer

- Pale Ales  $\leftarrow$  Bitters  
India Pale Ales  
Golden Ale  
Australian Sparkling Ale
- Brow ales  $\leftarrow$  Scottish Ale  
Mile & Brown Ale
- Dark Ales  $\leftarrow$  Porter  
Stout
- High Gravity October Ales  $\leftarrow$  Old Ale  
Burley wine

## Beer appearance

Colour (Browning)

- ① Caramelisation (160°C)  
\* removal of water
- ② Pyrolysis (230-250°C)  
\* bitter / astringent
- ③ Maillard  
\* heat sugar & amino acids

\* colour measurement  
SRM / EBC / Degrees Lovibond

## Hop additions

for **bittering** (60分前加)  
◦ Columbus / Pride of Ringwood  
◦ Cheaper hops

for **Aroma** (0-10分前加)  
◦ Cascade / Fuggle / Golding  
◦ Steep / hop back / Saaz  
◦ Premium

## Dual Purpose

▷ Ella / East Kent Golding

## Beer of Germany & Czech Republic

### Hop varieties

- Hallertauer Mittelfrüh (Bavarian)
- Saaz (Czech)
- Tettnanger
- Spalt (Bavarian)
- Hersbrucker
- Perle

example of globalisation is **AB InBev**

### Big Beer

- 起家:
1. production efficiency (economic of scale)
  2. distribution efficiency (bargain power)
  3. marketing

发展中:

- Transport innovation (road, truck)
- Packaging innovation (cans)
- Refrigeration (commercial and domestic)
- Media innovation (television)
- Brewing innovation (industrial lager)

### Malts

**Vienna** & **Pilsner** & **Munich** malts

- ▷ high diastatic power
- ▷ low level of browning (美拉德)
- ▷ higher DMS than British
- ▷ less well modified than British which leads to more malty flavour

### Business model

商业模式:

- Brew lager, at scale  
最大规模生产
- lock up supply and distribution (own pubs and demand exclusively)
- Build brand loyalty
- respond to demographic and sociocultural change.

## History of Beer

### Sumerians (Iran)

- \* First recorded brewer
- around 3000 BC
- Beer - "Kars"

### Ancient Egypt

- drink beer to give thanks to gods

South VS North  
↓ wine Beer

### Middle ages

- 500-1000 AD, the first half of middle ages, brewing begins to be practiced in Europe.
- Shift from family tradition to centralised production

## Alcohol responsible consumption

### Alcohol detoxification in liver

Alcohol → Acetaldehyde → Acetic acid  
↓  
Acetyl-CoA

Guidelines: Adults < 10 standard drinks a week  
< 4 standard drinks a day

2: no alcohol for children

3: pregnant no alcohol

### Hopped beer

- First appear in Bremen (part of German Hanseatic League) around 1000 AD
- Hopped beer known as "white beer"
- Hop beer arrive in England about 1500
- By 1600, brewing is largely in German, Flanders, Netherlands and UK.

### Standard drink

- contains 10g of pure alcohol

再算 1 pot (285ml) of a 5% ABV beer  
 $285 \times \frac{3.95}{100} = 11.25 \text{g Alcohol}$   
 $\approx 1.1 \text{ standard drinks}$

换算 ABV → ABW

5% ABV drink =  $5 \times 0.79 = 3.95\%$  Alcohol by weight (ABW)  
 天 0.79

## Belgium Beers

### Belgium beer style

- A Ale
  - Witbier / Weizenbier
  - Belgian strong golden Ale
  - Blond
  - Dubbel
  - Tripel
  - Belgian Strong Dark Ale
  - Saison [farmhouse ale]

Commercial sour beer  
 • Lambic  
 • Berliner Weiss  
 • Gose

### Blending beer why?

- \* Add desirable characteristics
- \* dilute undesirable characteristics
- \* Maintain consistency across batches

## History of beer style

Porter: 1722

Stout: 1667

Pale ale: early 18th century

'Burton' Ale: 1600,

### History of Lager

- 1830, cold ferment is achieved by ice caves
- 1842, first golden lager in Plzen, Bohemia
- Commercial refrigeration 1860 in US.

### Alcohol free drink

- < 0.5% ABV
- ① Dilution
- ② Dealcoholisation
  - distillation 蒸馏
  - Membrane filtration
  - vacuum distillation
- ③ Limit fermentation
  - special yeast
  - cold contact fermentation
  - Limit sugars

### Abbey beer

- In many countries, brewing centre around monasteries
- Abbey beer a product in a similar style to a monastic beer

### Trappist beer

- beer must be brewed in the monastery

### Candi sugar 甜菜糖

- provide colour
- flavour compounds (melanoidins)
- increase gravity

### Wild Yeasts

- fruit are good source
- Not domesticated
- unpredictable in nature

Genus of wild yeast

Lactobacillus spp  
 Pediococcus spp

## Lactic Acid Bacteria (LAB)

乳酸菌

Sugar

→ Energy

Yeasts

Lactic Acid  
 - Oxygen  
 Lactic fermentation

Pyruvate

- Oxygen  
 → Alcohol & CO<sub>2</sub>  
 Alcoholic fermentation

+ Oxygen  
 CO<sub>2</sub>, H<sub>2</sub>O & Energy

### Craft beer

- smaller scale
- independent ownership
- greater diversity of products

### Business model 商业模式

#### A Owning brewery

- clear location but expensive

- food margin customer

- Use contact facilities

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'test the water' but legitimacy

758 question

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